M4: Quizzes 1 - covers Modules 1, 2 and 3

Due Fri 7 Jul 11:59am | **Points** 200 | **Questions** 11
Available Fri 7 Jul 12:01am - Sun 9 Jul 11:59pm | **Time Limit** 60 Minutes

Instructions

This is a timed quiz covering material covered in our recordings of the lectures covered in Modules 1, 2 and 3. There are 20 questions in total, worth 200 points. It is recommended that you complete all other activities in these Modules before you attempt this quiz. Some questions require you to use the tools or software you are allowed.

Remember that you can use the Discussion Board to ask questions, or contact me if you are stuck for longer than a week. You also have a forum in Ed Lesson.

Once you begin this quiz, you will have 60 minutes to complete it. Your first attempt, for marks, will be prepared by the system, and will be based on the most recent version of the quiz that you have seen. Subsequent attempts will be based on the most recent version of the quiz available to you.

Complete the quiz by the given due date. Late, open-book tutorials will not be accepted.

This quiz was released for Fri at 12:01pm.

Attempt History

	Attempt	Time	Score	Regraded
Latest	Attempt 1	10 minutes	175.22 out of 200	Not yet graded

Score for this quiz: **175.22** out of 200
Submitted Jul 7 at 12:10pm
This attempt took 10 minutes.

Question 1 — 5 / 5 pts

The following symbolic formula is a tautology:

"Pierce is a large triangle"

Correct!

- True
- False
- Pierce is a large triangle is a tautology. It is a proposition.
- None

Question 2 — 5 / 5 pts

Assuming:

$p = \text{true}$
$q = \text{false}$
$r = \text{true}$

What is the Boolean value of the expression:

$\neg r \vee (p \wedge q)$

Hint: Remember operator precedence!

Correct!

- The answer is true.
- The answer is false.
- True
- False

Question 3 — 5 / 5 pts

Assuming p is true and q is false. What is the value of:

$(p \vee q) \wedge (\neg p \vee q)$

Correct!

- The answer is true.
- The answer is false.
- True
- False
- The answer is false.

Question 4 — 5 / 5 pts

Identify the error in the truth table(s) of the following truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Incorrect!

- Neither the truth table nor the error!
- Neither the truth table nor the error!

Correct Answer

- The error in the truth table!
- The error in the truth table!

Question 5 — 5 / 5 pts

Which of the following is the **converse** of:

"If a gets, I expect it"

- If we don't get, I expect
- If we don't get, I expect it
- If I expect it, I get it
- If I don't expect it, I don't get it

Incorrect!

Correct Answer

- If I don't expect it, I don't get it

Question 6 — 5 / 5 pts

Which of the following is the **contrapositive** of:

"If the sales crashed, the CEO will be sacked."

Correct!

- If the CEO is sacked, the sales crashed.
- If the CEO is not sacked, the sales crashed.
- If the sales crashed, the CEO is sacked.
- If the CEO is not sacked, the sales crashed.

Question 7 — 5 / 5 pts

Which of the following is the **inverse** of:

"If the sales are good, I am a rich man."

- If I am not a rich man, the Sales are not good.
- If I am not a rich man, the Sales are good.
- I am a rich man.
- If the sales are good, I am a rich man.
- I am a rich man.

Question 8 — 5 / 5 pts

Which of the following is the **contrapositive** of:

$q \vee r \rightarrow p$

Correct!

- $\neg(q \vee r) \rightarrow \neg p$
- $p \rightarrow (q \vee r)$
- $\neg p \rightarrow \neg(q \vee r)$
- $p \rightarrow (q \wedge r)$

Question 9 — 5 / 5 pts

If the proposition

$p \rightarrow q$

is false, which of the following must also be true? Select all that apply.

- $\neg p \rightarrow \neg q$
- $p \rightarrow q$
- $q \rightarrow p$
- $\neg q \rightarrow \neg p$

Correct!

Question 10 — 5 / 5 pts

Select all of the following statements that are **logically equivalent** to:

$\neg(p \rightarrow q) \rightarrow q$

Correct Answer

- $\neg q \rightarrow (p \vee q)$

Correct!

- $p \wedge \neg q \rightarrow q$
- $\neg q \rightarrow (p \wedge q)$
- $\neg q \rightarrow (p \vee q)$
- $(p \vee q) \rightarrow q$

Question 11 — 5 / 5 pts

In the following proof, indicate which line of propositional logic was used to reach the indicated step. For example, on step 2 you should indicate what law was used to get from the step 1 to step 2.

Fill in the blanks following propositions:

$p \rightarrow (p \vee (r \wedge q))$
$(p \rightarrow (p \vee (r \wedge q)) \wedge (